

SAVONLINNA PUNKAHARJU, Finland

WMO: 027780

Lat: 61.8019N

Lon: 29.3153E

Elev: 80

StdP: 100.37

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.3	(c) -22.9	(d) -29.1	(e) 0.3	(f) -26.3	(g) -25.2	(h) 0.4	(i) -22.8	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.5	(c) 27.3	(d) 19.1	(e) 25.2	(f) 18.2	(g) 23.2	(h) 17.0	(i) 20.5	(j) 25.2	(k) 19.3	(l) 23.4	(m) 18.2	(n) 22.1	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 18.9	(b) 13.8	(c) 22.4	(d) 17.8	(e) 12.9	(f) 21.5	(g) 16.5	(h) 11.9	(i) 20.0	(j) 59.3	(k) 25.2	(l) 55.4	(m) 23.5	(n) 51.6	(o) 22.2	(p) 24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -27.3	(f) 28.8	(g) 4.8	(h) 2.5	(i) -30.8	(j) 30.6	(k) -33.6	(l) 32.1	(m) -36.3	(n) 33.5	(o) -39.8	(p) 35.4		
(a) N/A	(b) N/A	(c) N/A	(e) -27.5	(f) 21.6	(g) 4.8	(h) 1.8	(i) -30.9	(j) 22.9	(k) -33.7	(l) 23.9	(m) -36.4	(n) 24.9	(o) -39.9	(p) 26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-9.1	-7.1	-3.2	3.0	10.0	14.2	17.4	15.7	10.9	4.2	0.2	-4.3
	DBStd	9.99	7.49	6.86	4.89	3.54	4.34	3.76	3.44	3.08	3.16	3.44	4.01	6.02
	HDD10.0	2696	591	478	409	211	54	8	0	1	27	181	294	442
	HDD18.3	5137	849	711	668	461	258	133	59	93	222	438	544	701
	CDD10.0	652	0	0	0	1	55	133	230	178	55	1	0	0
	CDD18.3	51	0	0	0	0	1	8	31	11	0	0	0	0
	CDH23.3	406	0	0	0	0	26	74	231	75	0	0	0	0
	CDH26.7	77	0	0	0	0	2	12	51	13	0	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	618	45	37	33	28	41	62	69	75	59	61	55	50
	PrecMax	744	73	73	58	74	114	99	147	135	164	130	104	83
	PrecMin	477	13	5	6	7	7	19	24	19	20	15	10	20
	PrecStd	75	17	17	15	14	25	20	32	38	33	27	22	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	4.6	9.2	17.6	26.2	28.2	30.3	28.4	20.4	13.2	9.8	6.4
		MCWB	2.3	2.4	3.9	9.7	15.8	17.7	21.0	20.5	15.9	11.5	8.5	5.7
	2%	DB	2.1	2.7	6.0	13.9	23.3	25.4	28.1	25.4	18.3	11.4	7.6	4.2
		MCWB	1.6	1.7	2.8	7.8	14.2	17.6	20.0	18.6	14.8	10.1	6.9	3.4
	5%	DB	1.0	1.6	4.1	11.3	20.6	23.2	26.0	23.1	16.8	10.4	6.5	2.9
		MCWB	0.5	0.9	1.6	6.0	12.9	16.2	19.3	17.4	14.0	9.3	5.8	2.3
	10%	DB	0.2	0.8	2.7	9.1	18.1	21.2	23.8	21.2	15.7	9.3	5.1	1.7
		MCWB	-0.2	0.4	1.0	5.0	11.3	15.1	18.0	16.7	13.5	8.2	4.6	1.3
	0.4%	WB	2.6	3.0	4.7	10.4	17.7	20.6	22.5	21.8	16.7	11.6	8.7	5.7
		MCDB	2.8	4.0	7.6	15.7	23.8	25.7	27.6	27.2	19.1	12.4	9.4	6.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.7	1.9	3.2	8.5	15.4	18.8	21.1	19.5	15.7	10.6	6.9	3.7
		MCDB	2.0	2.7	5.2	13.1	20.5	23.1	26.4	23.4	17.4	11.2	7.4	4.2
	5%	WB	0.7	1.1	2.2	6.9	13.9	17.0	20.0	18.3	14.7	9.5	5.9	2.4
		MCDB	1.0	1.6	3.6	10.5	19.2	21.6	24.2	22.0	16.5	10.1	6.5	2.8
	10%	WB	0.0	0.5	1.3	5.5	12.4	15.7	19.0	17.2	13.8	8.3	4.7	1.4
		MCDB	0.4	0.9	2.4	8.4	17.4	20.1	22.9	20.4	15.2	9.2	5.1	1.7
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.1	7.9	9.1	10.3	9.1	8.5	7.8	6.1	4.7	3.3	4.0
		MCDBR	3.2	5.0	9.0	12.9	14.4	12.3	11.7	11.0	8.2	5.9	4.5	3.8
		MCWBR	3.2	4.4	6.2	7.6	6.8	5.7	4.8	5.1	5.0	4.7	4.3	3.6
	5% WB	MCDDBR	3.1	4.6	7.5	11.3	12.0	11.1	10.4	9.6	7.1	5.2	4.4	3.6
		MCWBR	3.2	4.1	5.6	7.0	6.6	5.7	4.8	4.8	4.8	4.7	4.2	3.6
(40) Clear-Sky Solar Irradiance	taub		0.246	0.294	0.320	0.359	0.355	0.351	0.376	0.361	0.338	0.319	0.265	N/A
	taud		2.039	2.116	2.193	2.337	2.444	2.486	2.431	2.457	2.510	2.401	2.127	N/A
	Ebn at Noon		433	652	783	829	867	877	839	823	778	644	407	N/A
	Edh at Noon		38	71	94	101	99	96	100	90	71	55	34	N/A
	RadAvg		0.22	0.78	2.26	3.65	4.95	5.30	5.12	3.93	2.24	0.89	0.24	0.11
(45) All-Sky Solar Radiation	RadStd		0.04	0.19	0.32	0.41	0.58	0.32	0.48	0.47	0.26	0.13	0.04	0.01

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page